Decentralized Graph-of-Thoughts with OSM Map Grounding for Cooperative Autonomous Driving

Ilyas Faresse (329743), Imade Bouhamria (339659), Paul Tercier (347407), Adrian Martínez López (396379)
CS-503 Final Report

***Abstract*—The V2V-GoT approach formulates the task of cooperative autonomous driving as a Graph-of-Thoughts (GoT) among nine question nodes, yet with a centralized large-language model (LLM) that fuses the raw data from LiDAR scans obtained from all connected autonomous vehicles (CAVs). In our study, we extend V2V-GoT in a decentralized way on two axes independently. First, using the metadata available in the V2V4Real dataset, we get the GPS trajectories of all vehicles and integrate the OSM map data of the local road network graph into the GoT pipeline through two methods: an Overpass API query for pure-text data and a ViT encoder processing satellite imagery. Second, we decentralize the pipeline, which implies that all questions Q1-Q9 are executed locally at each individual CAV, and instead of exchanging features across CAVs, we exchange plain text messages between CAVs. The result is that decentralized V2V-GoT performs equally to the original one with only a marginal drop in performance, namely -0.10 m L2 and -0.12% collisions on V2V4Real, yet reducing cross- CAV communication 1000 times to 0.0003 MB versus 0.4068 MB per frame. More interesting, combining the two axes leads to: stacking the textual OSM grounding over the decentralized chain proves more effective than the centralized setup, reducing the planning L2 error from 2.62 m to 1.80 m (-31%), reducing the prediction L2 error from 7.62 m to 5.86 m (-23%) and reducing the collision rate almost in half, from 1.84% to 0.98%, without adding even one penny in terms of the inter-CAV communication overhead.**

I. Introduction

V2V cooperative perception allows a connected vehicle to perceive further than its onboard sensor by communicating with other vehicles within its vicinity. The goal is simple: A car's LiDAR cannot see any objects occluded by a truck, a building, or a hill, while another vehicle nearby certainly can. Recently, research papers have posed cooperative driving as a question-answer problem using a multimodal LLM. V2V-LLM [1] pioneered this line of attack with a multimodal LLM on the V2V-QA dataset (created on the foundation of V2V4Real [2]) and V2V-GoT [3] expanded on it by posing a graph-of-thoughts chain on nine question nodes relating to occlusion-aware perception, planning-aware prediction, and planning. However, both frameworks rely on the same two assumptions: the *centralized-fusion* assumption (where one centralized LM takes the BEV LiDAR features of all the CAVs in one go) and the *map-blindness* assumption. Centralized approach suffers from a high bandwith overhead, tight time-synchronization constraint, and disregards the trust boundary among manufacturers. Both challenges are considered, and each research question corresponds to a single contribution.

RQ1: Will a concrete map prior decrease road-geometry violation? Most of the errors in the V2V-GoT plans are caused by road geometry violations and not by the perception module. We extract per-frame GPS coordinates from V2V4Real metadata, match it to LiDAR BEV frame and experiment with two different types of local OSM textual query with Overpass API, OSM image provided to the visual encoder and ViT encoder with lightweight LoRA fine-tuning.

RQ2: Will full decentralization of the V2V cooperation deteriorate planning? We perform all nine steps of the question-and-answer chain on per-CAV basis, where cross-CAV feature fusion is replaced with compact text messages exchange or no message at all between question nodes: visible objects from the receiver's Q1 provided to Q3 of the senderm, sender's Q9 plan provided to Q6 of the receiver.

The two contributions complement each other, with OSM injection working independently of the decentralization step. The results will be provided for all nine test scenarios from V2V4Real (1,723 frame pairs total, 3,446 samples).

II. Related Work

Backbones for cooperative perception. CoBEVT [4], Where2comm [5], and OPV2V [6] investigate multi-CAV feature-level fusion without utilizing an LLM. Their detector backbone is adopted in V2V-GoT while our attention is on reasoning above.

Planning with map conditionings. The work by TOKEN [7] demonstrates that HD maps are essential to produce better trajectories in nuScenes. HD maps are not available in many places around the world and may even be owned by an operator company. Thus, they are replaced with OpenStreetMaps acquired via the public Overpass API.

V2V4Real dataset. V2V4Real [2] is the source real-world dataset containing sequences captured using two connected cars each mounted with a LiDAR sensor on the roof. Both V2V-QA

Vehicle-to-vehicle (V2V) cooperative perception enables a connected car to have perception beyond its sensors using data exchange with other nearby vehicles. The need is clear: a vehicle's own LiDAR cannot perceive any object hidden behind another vehicle, structure, or hill, but a nearby vehicle frequently can. In recent research, cooperative driving was posed as a question answering task on top of multimodal LLM. V2V-LLM [1] laid down the groundwork of this approach by developing V2V-QA on V2V4Real [2], which was later expanded to V2V-GoT [3] with a graph-of-thoughts framework across nine question nodes for occlusion-aware perception, planning-aware prediction, and planning. Both approaches make a *centralized-fusion* assumption (an LM consumes BEV LiDAR features from all CAVs in one pass) and stay *map-blind* (neither road geometry nor map is known to the planner). Centralization requires high bandwidth, time synchronization, and doesn't take into account the security issue of cooperation between vehicles built by different brands, map blindness means incorrect lane trajectory generated by the planner even if perception is accurate.

III. Method

We extend the V2V-GoT pipeline without modification, and apply two independent techniques: map priors injected into selected GoT models, and a decentralization scheme which removes cross-CAV feature fusion in favor of text messaging.

A. GPS Recovery and OSM Alignment

Two complementary injection approaches are considered. For the first one, a *text-only prior* is proposed, that involves querying Overpass on the GPS-recovered coordinates corresponding to the road segment below the ego vehicle, converting the response to a minimalistic JSON with three elements (lanes number, textual description, confidence), and adding this JSON object to the LLaVA prompt as extra tokens. This approach does not require retraining of any model parameters, is computationally cheap, and can be used to augment any GoT node arbitrarily.

The other injection strategy uses a *ViT visual prior*. A satellite image cropped from GPS-recovered coordinates is injected into a novel ViT encoder with accompanying modality projector mapping its outputs into LLaVA feature space. This module works in parallel with the original feature extraction pipeline for LiDAR data. In order to avoid the backbone treating satellite tokens as noise, we fine-tuned LLaVA using light LoRA with the satellite injection module being trained from scratch.

B. OSM Map Grounding

Two different approaches are explored. The first one is called *text-only prior*: using the recovered GPS coordinates, we make an Overpass query regarding the road under the ego vehicle, extract three parameters (number of lanes, free-text description, and confidence score) and add them to the prompt in JSON format. This method is fully zero-shot, with no modifications needed, only an additional database query, allowing us to augment any GoT node as needed.

The second method uses *ViT visual prior*, where the same satellite image cropped with the recovered GPS coordinates is passed to a newly introduced ViT encoder and a modality-specific projective, which projects the output into the embedding space of the language model. The new branch runs parallel to the original LiDAR path and uses the same projector on the LiDAR side. To prevent backbone treating satellite tokens as noise, we perform lightweight LoRA finetuning of LLaVA, while training the ViT projector from scratch.

The cheap text-only prior loses the majority of the spatial information captured in the satellite tile; the ViT prior uses the entire tile but is the sole approach which must undergo training. We apply the ViT prior at Q9 (the node of the trajectory) where visual tokens are prevalent and the text prior across all nine nodes in an ablation exercise. One other technique that was explored during development but ultimately abandoned was the use of a rasterized OpenStreetMap image directly as LLaVA's visual context since it was obviated by the ViT encoder path which provides greater flexibility in the projector.

Before committing any GPU resources to either path, we tested the ability of the model to accept a new modality by using a synthetic road representation: the controlled map-like channel provided evidence that the projector and backbone can accept a new modality without collapsing, and the map data is able to affect planning.

C. Decentralization

The original V2V-GoT formulation assumes a centralized inference pipeline in which all CAVs transmit their local observations to a central LLM. In practice, each CAV sends its own LiDAR representation together with the textual context accumulated along the question-answer chain.

While this design maximizes the amount of information available to the planner, it introduces a major scalability limitation. In a realistic deployment, every CAV

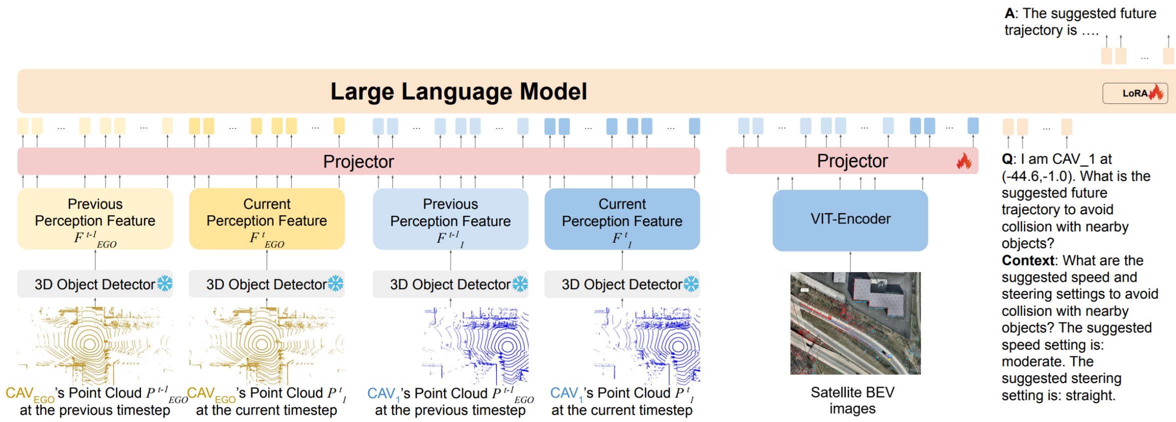

Figure 1: TODO. **Figure adapted from Chiu et al. [3].**

would need to continuously broadcast high-dimensional LiDAR features to neighboring vehicles or to a centralized server. For a system composed of N vehicles, this results in an all-to-all communication pattern with $\mathcal{O}(N(N-1))$ exchanged messages per timestep. Since LiDAR feature tensors are significantly larger than text messages (around 0.4MB when compressed), the resulting bandwidth and synchronization requirements rapidly become impractical for real-world deployment.

This motivates the exploration of decentralized cooperative reasoning, where each CAV performs inference independently and cooperation is restricted to lightweight textual communication. Our objective is therefore to quantify how much planning and perception performance deteriorates when raw sensor sharing is removed, and to determine whether text-based coordination can compensate for the missing information.

We evaluate two decentralized settings.

Fully isolated decentralization. As a first baseline, we completely isolate both CAVs during inference. Each vehicle executes the full Q1–Q9 Graph-of-Thought chain locally using only its own LiDAR observations and internal reasoning state. No LiDAR features, intermediate embeddings, or textual messages are exchanged between agents. This setting establishes the lower-bound performance achievable without any cooperation and allows us to directly measure the contribution of inter-agent information sharing in the original centralized formulation.

Text-based decentralized cooperation. After establishing the isolated baseline, we progressively reintroduce cooperation using lightweight text messaging between CAVs. Instead of exchanging raw LiDAR features, agents communicate compact natural-language summaries derived from intermediate GoT nodes. The communication protocol follows the semantics already present in the original V2V-GoT reasoning graph, where outputs from upstream nodes can be reused as contextual information for downstream planning and prediction tasks.

TODO: explain here the ideas that we have tried q1 from cav1 to q3 from cav 0 - nice result enrich q9 - went wrong finetuning of q9 - went wrong

IV. Experiments and Results

A. Experimental Setup

All experiments are conducted on the V2V4Real validation split following the same evaluation protocol used in V2V-GoT [3]. We reuse the original question-wise metrics for all Q1–Q9 nodes, including perception F1 scores, trajectory L2 distance, action accuracy, steering/speed prediction error, and collision rate.

We evaluate the following inference variants: centralized V2V-GoT, fully decentralized inference without communication, lightweight text-based messaging schemes, and the proposed OSM-grounded extensions. Communication overhead is additionally measured as the average transmitted payload per frame between CAVs.

B. Results

Table I reports every variant on the V2V4Real validation split.

The per-vehicle Q1–Q9 chain reasons about each CAV in isolation, so two locally sensible plans can still collide once executed jointly. To bound this, we add a deterministic resolver that runs on the two Q9 trajectories after inference. It is rule-based and applied identically to every variant in Table I, acting as a final safety layer rather than a learned module, so it can sit on top of any model without retraining.

Conflict definition. We first need a concrete trigger. Each car is represented by its oriented hitbox swept along its predicted Q9 trajectory, and we take the minimum hitbox-to-hitbox distance between the two cars over the prediction horizon. A scenario is flagged as a conflict when this clearance drops below 3 m: below that margin, small prediction errors are enough to turn two close paths into contact.

Resolution rules. Each scenario passes through a fixed cascade and stops at the first rule that applies:

1) *Agreement.* Both cars predict the same speed class and the same steering; the plans are treated as coordinated and executed unchanged.
2) *No physical conflict.* The minimum clearance is at least 3 m; the paths do not intersect and are executed unchanged.
3) *Speed reduction.* The two cars fall in different speed classes; the faster car drops one speed rung to widen the time-to-contact margin.
4) *OSM right-of-way.* A map is available and the two roads carry different priority; the lower-priority car yields.
5) *Tiebreak.* No earlier rule resolves the case; `ego` deterministically yields to `CAV-1`, again realized as a speed reduction.

Rule 4 requires per-road priority and is not exercised in the reported run, so the numbers below come from the four remaining rules. *Results.* Figure 2 traces all 1,723 validation scenarios through the cascade. Of these, 84 contain a sub-3 m clearance before resolution. Agreement covers 25 of them but leaves them unchanged, since matching action classes are assumed coordinated. The remaining 59 reach the geometric and tiebreak branches: speed reduction clears 26, the tiebreak path settles 3, and 30 cannot be opened past the margin. After the cascade 55 scenarios still violate the 3 m clearance, down from 84, matching the resolver-only row of Table II. The resolver on its own is therefore a modest filter; its value grows once peer intent is injected at Q8, which shrinks the conflict set it has to handle.

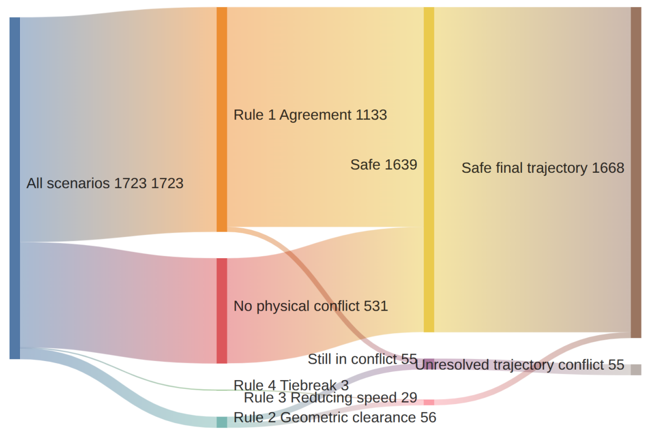

Figure 2: Deterministic conflict resolver. All 1,723 validation scenarios pass through the rule cascade; 84 start with a sub-3 m clearance and 55 remain unresolved after the cascade.

We tested whether peer information should be injected before the final resolver, and where in the V2V-GoT chain this information is most useful. The baseline runs both CAVs independently and only resolves conflicts afterward. Our variants inject a peer signal into either Q8, which predicts speed and steering, Q9, which generates the final trajectory, or both. We also compare full trajectory signals against compact alternatives.

The main trend is clear: peer information is most useful when injected into Q8, before speed and steering are fixed. Injecting the neighbor's full Q9 trajectory into Q8 reduces post-resolver conflicts from 55 to 7. Applying the same idea only after the resolver fails reaches the same final result, while being more compute-efficient because it reruns only the remaining hard cases.

Direct Q9 injection is less reliable. It helps when restricted to conflict frames, but becomes harmful when applied to all samples, increasing remaining conflicts to

Table I: V2V-GoT inference variants on V2V4Real val. Q1–Q4 are categorical metrics (F1, higher is better), Q5/Q7/Q9 are trajectory L2 in meters (lower is better); Q6 is action accuracy (%); Q8 is normalized L1 on the speed/steering tokens (lower is better); Q9 CR is the collision rate; Comm. MB is bytes transmitted between CAVs per frame.

Method	**Q1 F1↑**	**Q2 F1↑**	**Q3 F1↑**	**Q4 F1↑**	**Q5 L2↓**	**Q6 Acc↑**	**Q7 L2↓**	**Q8 L1↓**	**Q9 L2↓**	**Q9 CR↓**	**Comm. MB↓**
Central	**51.6**	30.1	43.8	60.7	8.06	87.5	7.62	0.0882	2.62	1.84	0.4068
Decent.	50.3	27.9	44.1	**62.3**	8.01	0.0	9.43	0.0897	2.85	2.28	**0.0000**
Q1msg	51.1	27.2	34.6	60.3	8.14	87.3	7.65	0.0879	2.72	1.96	0.0003
OSM-image (Q9 only)	51.6	30.1	43.8	60.7	8.06	87.5	7.62	0.0882	2.10	**0.81**	0.4068
OSM-text (All 9 Q-nodes)	51.6	30.1	43.8	60.7	8.05	87.5	7.64	0.0882	2.55	1.81	0.4068
OSM-text + Decent.	50.7	**30.5**	**49.4**	51.3	**6.18**	**90.8**	**5.86**	**0.0858**	**1.80**	0.98	**0.0000**

Table II: Pre-LLM peer-injection ablations on 1,723 validation frames. Conflict counts are frame counts; percentages are shown in parentheses.

Peer signal → target	Scope	Q8 speed acc.↑	Q8 steer acc.↑	Q9 endpoint err.↓	Pred. conflicts before resolver↓	Pred. conflicts after resolver↓	Conflicts if ego yields↓
None	all	**97.7**	**95.8**	8.869	84 (4.9%)	55 (3.2%)	62 (3.6%)
Q9-full → Q8	conflict-before	95.9	95.4	5.787	**14 (0.8%)**	**7 (0.4%)**	**11 (0.6%)**
Q9-full → Q9	conflict-before	**97.7**	**95.8**	6.115	22 (1.3%)	10 (0.6%)	17 (1.0%)
Q9-full → Q8+Q9	conflict-before	95.9	95.4	6.169	21 (1.2%)	14 (0.8%)	15 (0.9%)
Q9-full → Q8	all	84.7	88.8	8.128	16 (0.9%)	12 (0.7%)	12 (0.7%)
Q9-full → Q9	all	**97.7**	**95.8**	24.629	220 (12.8%)	168 (9.8%)	159 (9.2%)
Q9-full → Q8+Q9	all	84.7	88.8	24.749	207 (12.0%)	185 (10.7%)	155 (9.0%)
Q9-full → Q8	conflict-after	96.5	**95.8**	5.693	**14 (0.8%)**	**7 (0.4%)**	**11 (0.6%)**
Q9-endpoint → Q8	conflict-before	97.5	**95.8**	**5.637**	18 (1.0%)	9 (0.5%)	15 (0.9%)
Q7-compact → Q8	conflict-before	96.7	95.5	5.689	18 (1.0%)	8 (0.5%)	14 (0.8%)
Q6-compact → Q8	conflict-before	97.0	95.6	5.648	18 (1.0%)	9 (0.5%)	14 (0.8%)

168–185. Compact signals are also strong: Q7-compact leaves only 8 conflicts, while Q9-endpoint and Q6-compact leave 9. This suggests that the useful mechanism is not simply sharing more information, but sharing the right peer intent at the right node: Q8 should adapt the action choice, and the deterministic resolver should remain as the final safety layer.

C. Limitations

V2V4Real ground truth contains only 1 frame in 1,723 (0.06%) where the two CAVs' trajectories come within the resolver's 3 m threshold. This makes the conflict-resolution analysis useful as a stress test, but statistically limited as a benchmark for rare close-interaction cases.

A second limitation is compute availability. Several variants require repeated V2V-GoT inference, rerunning selected Q-nodes, or LoRA fine-tuning with additional OSM/satellite inputs. Since these experiments depend on shared GPU compute nodes, limited compute power and node availability constrained the number of full retraining runs, random seeds, and exhaustive ablations we could perform. The reported results should therefore be interpreted as the strongest validated configurations under the available compute budget, rather than as a complete search over all possible injection locations, prompts, and coordination policies.

V. Discussion

VI. Conclusion

Author Contributions

A.M.L. and I.F. conceived the two extension axes in discussion with all authors. I.F. recovered the true GPS coordinates from the V2V4Real metadata and validated alignment by overlaying LiDAR BEV onto the recovered positions, unblocking all downstream OSM work; he also implemented the OSM text-embedding injection and obtained the best-performing model by limiting per-node context through a combined decentralized and text-OSM configuration. A.M.L. and P.T. reproduced the V2V-GoT inference pipeline on V2V-GoT-QA (matching reported numbers within ±0.2%), characterized the per-epoch training cost on the available GPUs, built the project webpage, and developed the alternative decentralized variants. P.T. implemented the shared training pipeline and the synthetic road-representation sanity check for OSM-style token injection, developed the image-based OSM model, and investigated conflict-reduction strategies for the decentralized and q1msg settings under partial and no information sharing. I.B. explored alternative cooperative-driving datasets and adapted them toward the V2V-QA format, and studied late-stage information injection routing a CAV's final-node output back into another CAV's prediction Q-node. All authors discussed the results and contributed to the final manuscript.

Appendix A.
Architecture diagram

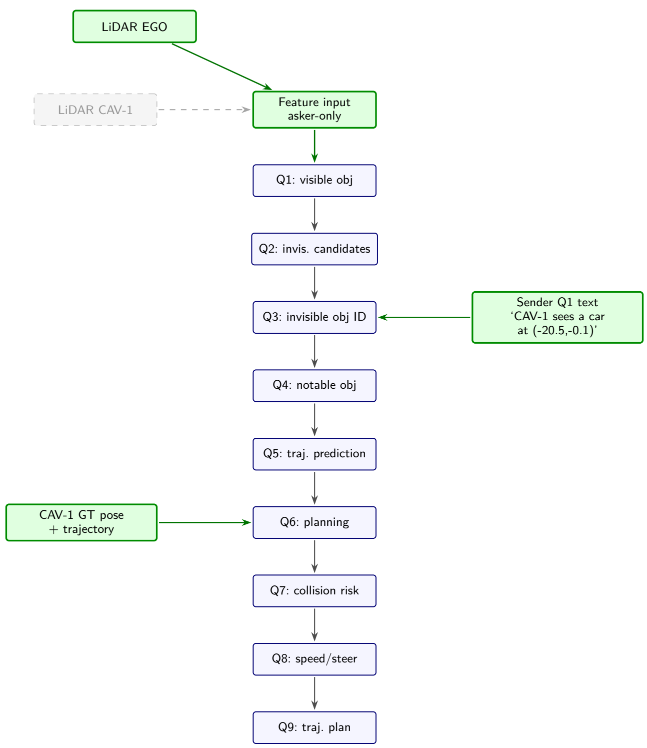

Figure 3: V2V-GoT q1msg inference flow

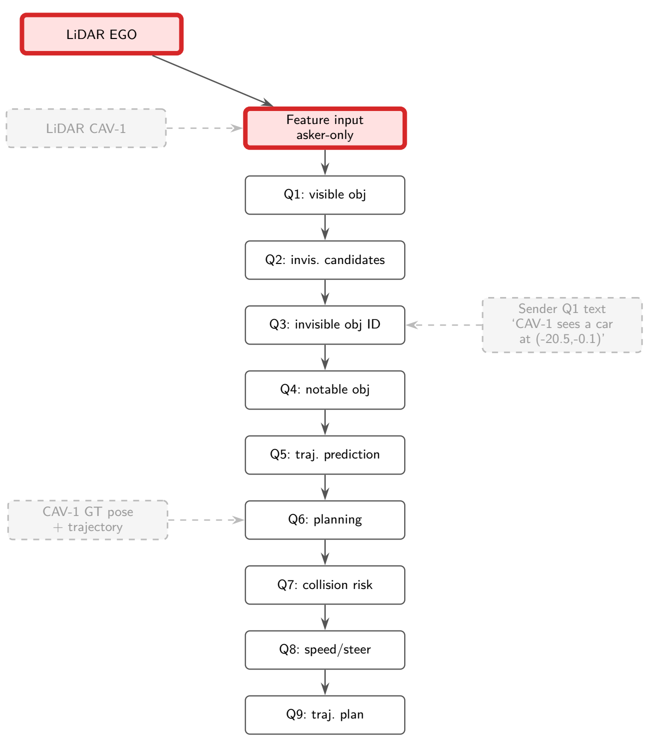

Figure 4: V2V-GoT decentralized inference flow

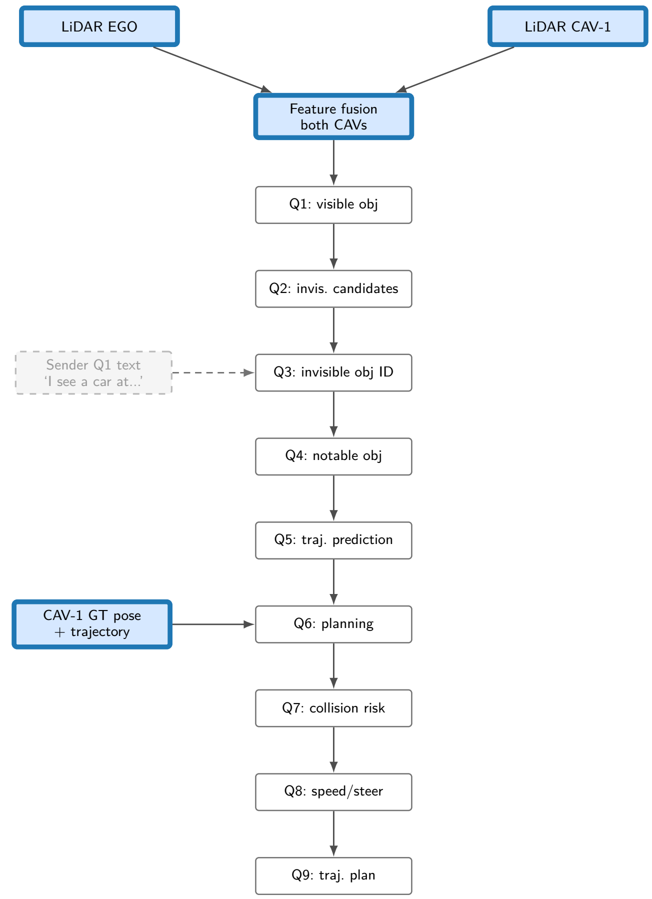

Figure 5: V2V-GoT centralized inference flow

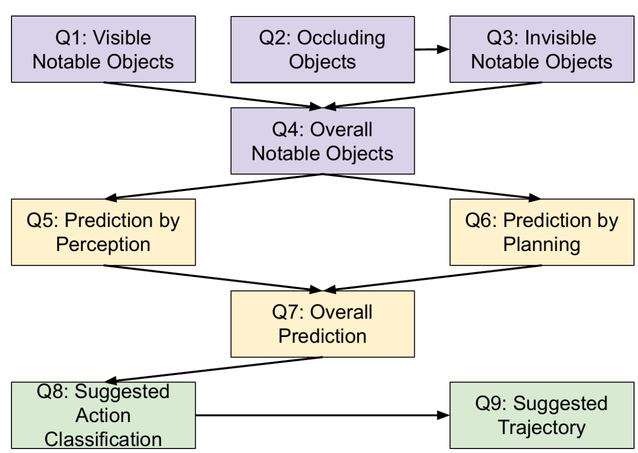

Figure 6: The V2V-GoT question graph. Perception nodes (Q1–Q3) feed the overall notable-object set Q4, which branches into prediction-by-perception (Q5) and prediction-by-planning (Q6); their fusion in Q7 drives action classification (Q8) and the final trajectory (Q9). **Reproduced from Chiu et al. [3].**

Appendix B.
Training Details

Appendix C.
RELATED WORK

A. V2V-LLM

V2V-LLM [1] is the first work to frame cooperative autonomous driving as an MLLM-based question-answering problem. It introduces the **V2V-QA** dataset, built on top of V2V4Real (and V2X-Real), with five types of natural-language QA pairs covering grounding at a reference location (Q1), grounding behind an occluder (Q2 and Q3), notable-object identification (Q4), and planning (Q5).

The accompanying V2V-LLM baseline is a single LLaVA-v1.5-7b MLLM in which a LiDAR-based 3D object detector (PointPillars) replaces the original CLIP image encoder. Each CAV extracts a scene-level feature map and object-level feature vectors from its own point cloud and ships them to a centralized LLM, which answers any CAV's driving question in one pass. The work compares this *LLM fusion* against no-fusion, early-fusion, and feature-level intermediate-fusion baselines (V2VNet, CoBEVT, V2X-ViT, AttFuse) on V2V-QA, and shows that V2V-LLM is best on notable-object identification and planning.

B. V2V-GoT

V2V-GoT [3] extends V2V-LLM with a graph-of-thoughts reasoning framework. The authors curate a new **V2V-GoT-QA** dataset, also built on V2V4Real [2], that expands the original five V2V-QA tasks into nine interconnected question nodes (Q1 . . . Q9) organised around two ideas: *occlusion-aware perception* (Q1–Q4) and *planning-aware prediction* (Q5–Q7), followed by planning (Q8–Q9). Different QA types are connected by directed edges in the graph; the answer of a parent node is used as the input context of its children. The chain is:

- **Q1** visible notable objects.
- **Q2** occluding objects.
- **Q3** invisible notable objects (hidden behind occluders), takes Q2.
- **Q4** overall notable objects, takes Q1 and Q3.
- **Q5** prediction by perception, takes Q4.
- **Q6** prediction by planning (other CAVs' planned trajectories), takes Q4.
- **Q7** overall prediction, takes Q5 and Q6.
- **Q8** suggested speed and steering (action class), takes Q7.
- **Q9** suggested trajectory, takes Q8.

The V2V-GoT model itself reuses V2V-LLM's LLaVA-v1.5-7b backbone (with a LiDAR-based detector replacing the image encoder), and additionally consumes perception features at both the current and the previous timestep to capture temporal dynamics. The MLLM is queried once per QA node, with the previous answer threaded in as context.

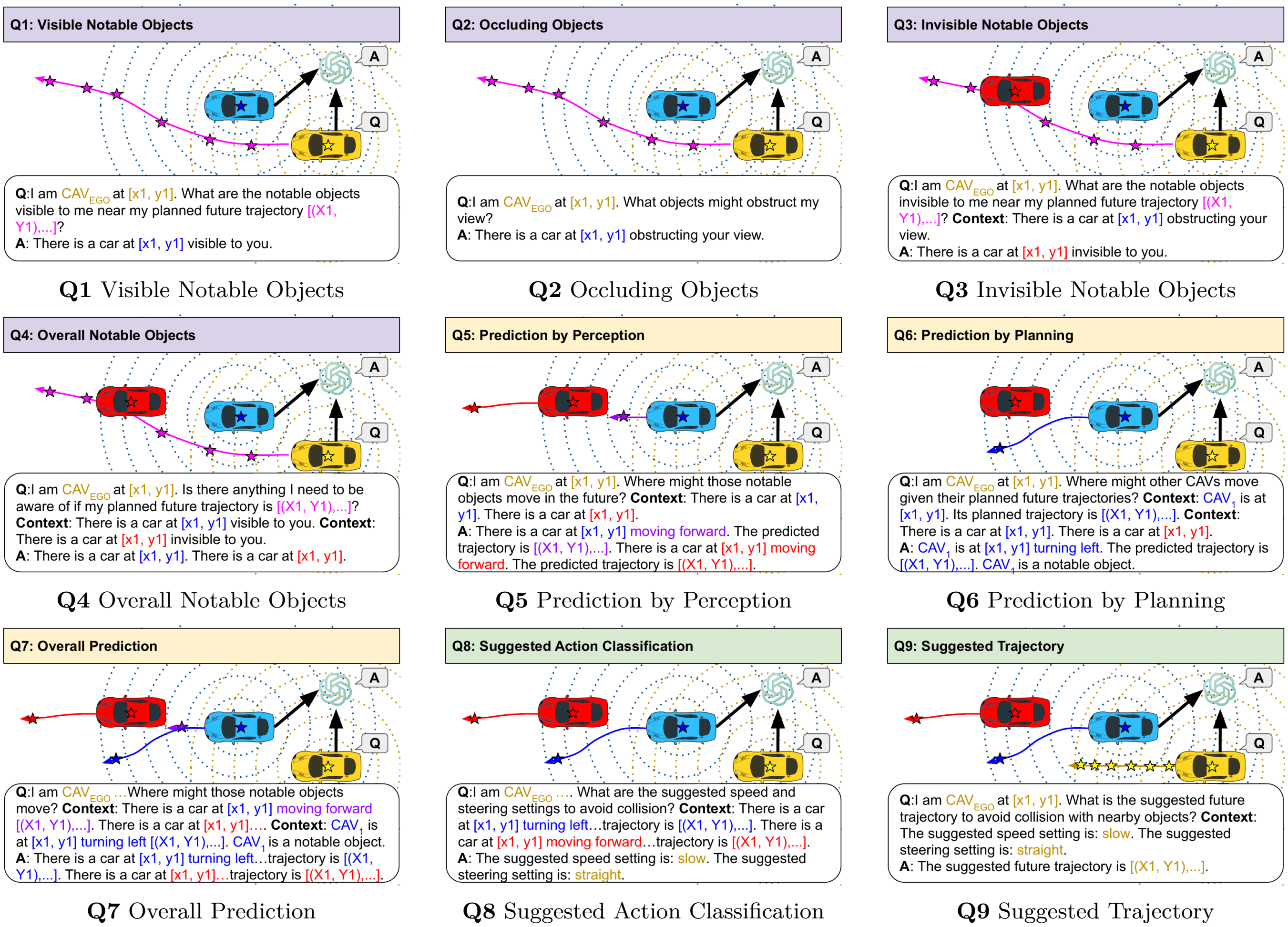

Figure 7: Illustration of V2V-GoT's nine QA pairs: perception (Q1–Q4), prediction (Q5–Q7), and planning (Q8–Q9). Black arrows pointing at the MLLM indicate perception data shared by the CAVs; other coloured arrows represent predicted or suggested future trajectories. **Reproduced from Chiu et al. [3].**

Appendix D. Additional Qualitative Results

Q1: visible notable objects

Table III: Q1 visible notable objects: per-metric performance across inference variants. Best in each row in **boldface**.

Metric	Central	Decent.	Q1msg
binary F1	**0.941**	0.869	0.936
binary precision	0.915	**0.927**	0.915
binary recall	**0.968**	0.818	0.959
loc F1 @ 0.5	0.254	**0.266**	0.256
loc F1 @ 1	0.337	**0.341**	0.335
loc F1 @ 2	0.394	0.399	**0.402**
loc F1 @ 4	**0.516**	0.503	0.511
parse error rate	**0.005**	0.008	0.009

Q2: occluding objects

Table IV: Q2 occluding objects.

Metric	Central	Decent.	Q1msg
binary F1	**1.000**	**1.000**	**1.000**
loc F1 @ 0.5	**0.129**	0.081	0.082
loc F1 @ 1	**0.177**	0.132	0.133
loc F1 @ 2	**0.224**	0.196	0.202
loc F1 @ 4	**0.301**	0.279	0.272
parse error rate	0.063	**0.044**	0.052

Q3: invisible notable objects

Table V: Q3 invisible notable objects. This node is the V2V messaging entry-point in the Q1msg variant.

Metric	Central	Decent.	Q1msg
binary F1	**0.513**	0.478	0.422
binary precision	**0.501**	0.462	0.365
binary recall	**0.525**	0.495	0.502
loc F1 @ 0.5	0.219	**0.265**	0.212
loc F1 @ 1	0.365	**0.414**	0.311
loc F1 @ 2	0.419	**0.440**	0.339
loc F1 @ 4	0.438	**0.441**	0.346

Q4: notable object identification

Table VI: Q4 notable object identification (Q1 $\cup$ Q3).

Metric	Central	Decent.	Q1msg
accuracy	**0.905**	0.806	0.893
F1 (overall)	**0.936**	0.859	0.929
precision	**0.911**	**0.911**	0.905
recall	**0.963**	0.814	0.954
F1 @ 0.5	0.270	**0.307**	0.274
F1 @ 1	0.368	**0.400**	0.364
F1 @ 2	0.436	**0.465**	0.433
F1 @ 4	0.607	**0.623**	0.603
avg dist error (m)	**5.88**	5.91	6.06

Q5: object motion prediction

Table VII: Q5 object motion prediction from perception.

Metric	Central	Decent.	Q1msg
action accuracy	0.705	**0.753**	0.743
L2 (m, avg)	8.06	**8.01**	8.14
binary F1	**0.939**	0.861	0.932
binary precision	**0.911**	**0.911**	0.906
binary recall	**0.969**	0.816	0.960
loc F1 @ 0.5	0.254	**0.268**	0.256
loc F1 @ 1	0.348	**0.351**	0.340
loc F1 @ 2	**0.411**	0.408	0.405
loc F1 @ 4	**0.529**	0.507	0.507

Q6: is the other CAV a notable object?

Table VIII: Q6 binary classification: is the other CAV a notable object?

Metric	Central	Decent.	Q1msg
binary class accuracy	**0.875**	0.000	0.873

Q7: motion prediction with full context

Table IX: Q7 motion prediction conditioned on Q5 and Q6.

Metric	Central	Decent.	Q1msg
action accuracy	0.706	0.723	**0.744**
L2 (m, avg)	**7.62**	9.43	7.65
binary F1	**0.939**	0.861	0.932
binary precision	**0.911**	**0.911**	0.906
binary recall	**0.969**	0.816	0.960
loc F1 @ 0.5	0.254	0.252	**0.256**
loc F1 @ 1	**0.347**	0.331	0.340
loc F1 @ 2	**0.411**	0.384	0.405
loc F1 @ 4	**0.524**	0.491	0.503

Q8: suggested speed and steering

Table X: Q8 suggested action class (speed and steering).

Metric	Central	Decent.	Q1msg
action accuracy	**0.918**	0.915	**0.918**
L1 (action edit dist.)	0.0882	0.0897	**0.0879**

Q9: suggested future trajectory

Table XI: Q9 planning: L2 and collision rate at 1/2/3 s, plus the NAVSIM PDMS decomposition (NC, TTC, C, PDMS).

Metric	Central	Decent.	Q1msg
L2 error @ 1s (m)	**1.645**	1.784	1.695
L2 error @ 2s (m)	**2.629**	2.859	2.729
L2 error @ 3s (m)	**3.590**	3.917	3.746
L2 error avg (m)	**2.621**	2.853	2.723
collision @ 1s	0.0012	0.0023	**0.0009**
collision @ 2s	0.0192	0.0218	**0.0177**
collision @ 3s	**0.0348**	0.0444	0.0403
collision avg	**0.0184**	0.0228	0.0196
NC (no collision)	**0.965**	0.956	0.960
TTC	0.998	**0.999**	**0.999**
C (comfort)	**0.683**	0.673	0.645
PDMS	**0.876**	0.865	0.861
PDMS sample avg	**0.877**	0.867	0.862

References

[1] H.-k. Chiu, B. Ivanovic, M. Pavone *et al.*, “V2V-LLM: Vehicle-to-vehicle cooperative autonomous driving with multi-modal large language models,” *arXiv preprint arXiv:2502.09980*, 2025.

[2] R. Xu, X. Xia, J. Li, H. Li, S. Zhang, Z. Tu, Z. Meng, H. Xiang, X. Dong, R. Song, H. Yu, B. Zhou, and J. Ma, “V2v4real: A real-world large-scale dataset for vehicle-to-vehicle cooperative perception,” 2023. [Online]. Available: https://arxiv.org/abs/2303.07601

[3] H.-k. Chiu *et al.*, “V2V-GoT: Graph-of-thoughts reasoning for cooperative autonomous driving,” *arXiv preprint arXiv:2509.18053*, 2025.

[4] R. Xu, Z. Tu, H. Xiang, W. Shao, B. Zhou, and J. Ma, “Cobevt: Cooperative bird’s eye view semantic segmentation with sparse transformers,” 2022. [Online]. Available: https://arxiv.org/abs/2207.02202

[5] Y. Hu, S. Fang, Z. Lei, Y. Zhong, and S. Chen, “Where2comm: Communication-efficient collaborative perception via spatial confidence maps,” 2022. [Online]. Available: https://arxiv.org/abs/2209.12836

[6] R. Xu, H. Xiang, X. Xia, X. Han, J. Li, and J. Ma, “Opv2v: An open benchmark dataset and fusion pipeline for perception with vehicle-to-vehicle communication,” 2022. [Online]. Available: https://arxiv.org/abs/2109.07644

[7] R. Tian, B. Li, X. Weng, Y. Chen, E. Schmerling, Y. Wang, B. Ivanovic, and M. Pavone, “Tokenize the world into object-level knowledge to address long-tail events in autonomous driving,” *arXiv preprint arXiv:2407.00959*, 2024.